

Using “Wave” to understand spatial pattern and associated functions using spatial omics data

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5/28/26



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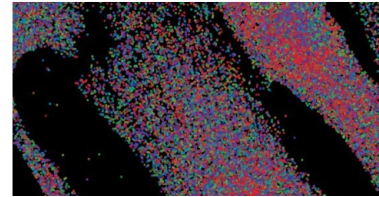
Spatial omics impact national and international level research

[nature](#) > [nature methods](#) > [focus](#)

Focus | 06 January 2021

Method of the Year 2020: spatially resolved transcriptomics

Spatially resolved transcriptomics is our Method of the Year 2020, for its ability to provide valuable insights into the biology of cells and tissues while retaining information about spatial context.

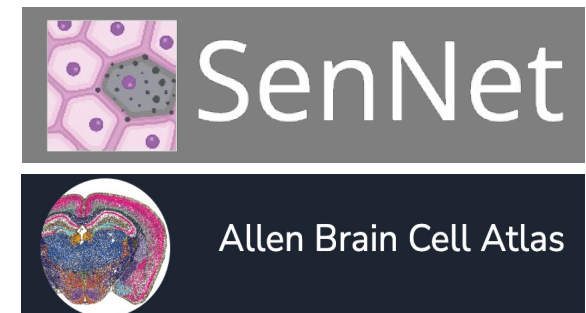
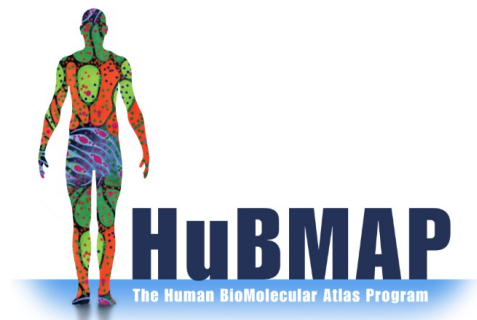


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Focus | 10 December 2024

Method of the Year 2024: spatial proteomics

Spatial proteomics is our pick for Method of the Year 2024, for the impact that these technologies have had on the understanding of the organization, structure and function of complex tissues, including in global tissue atlas projects.



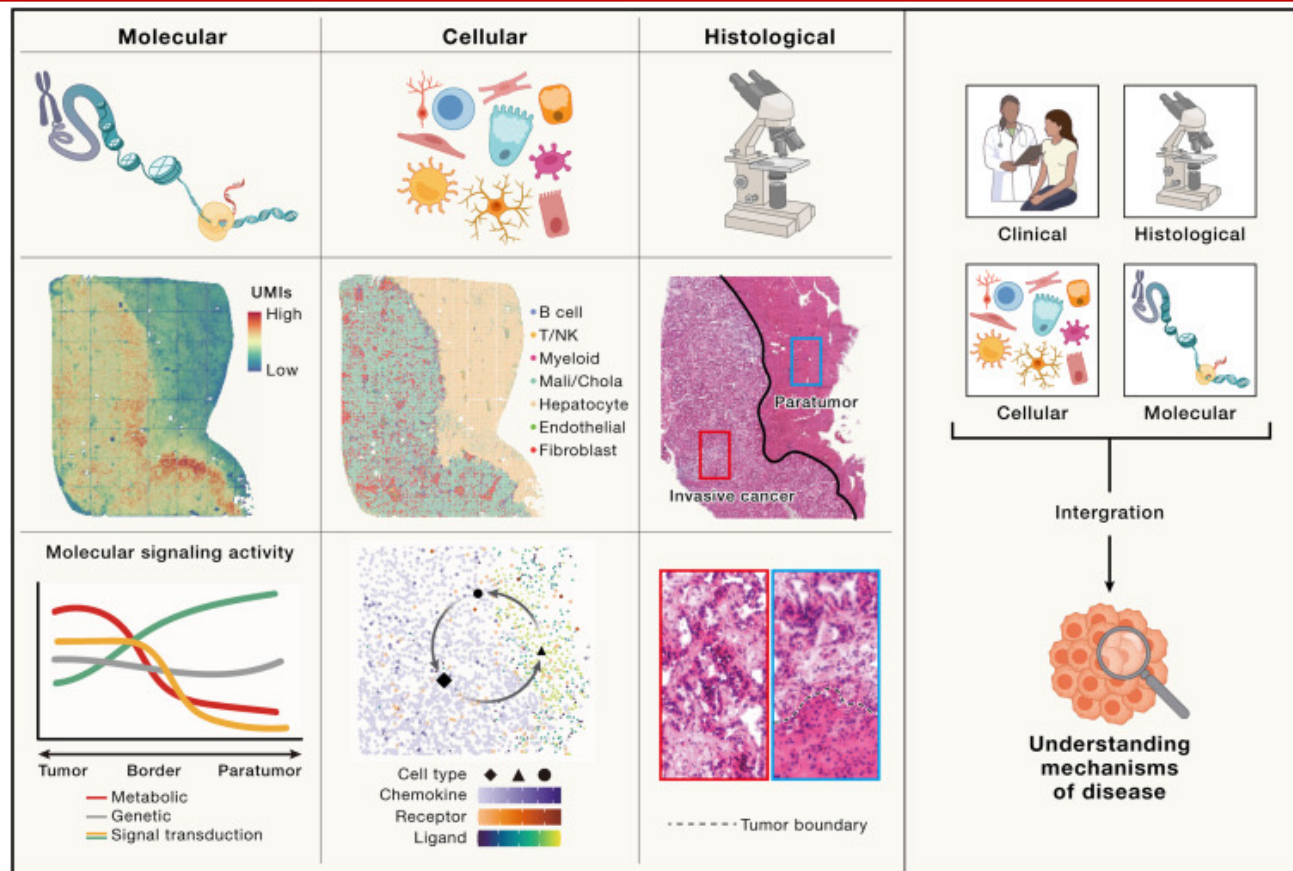
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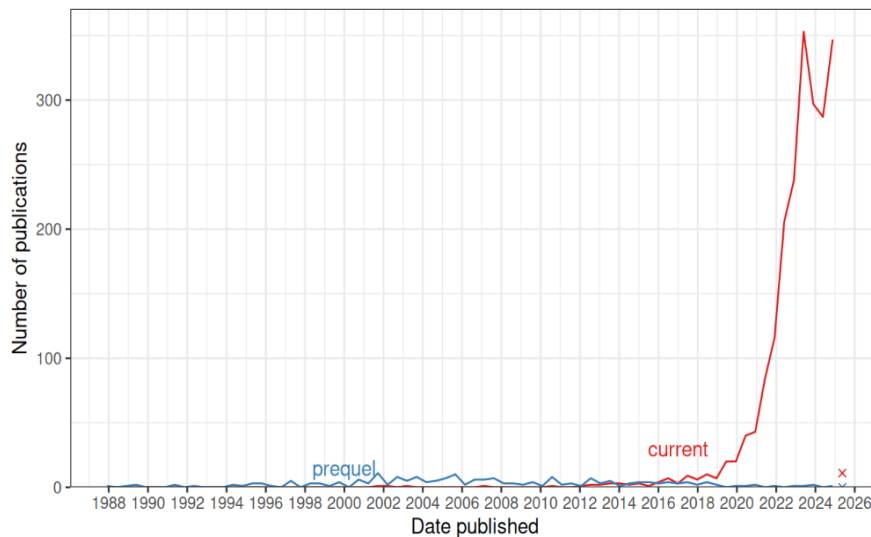


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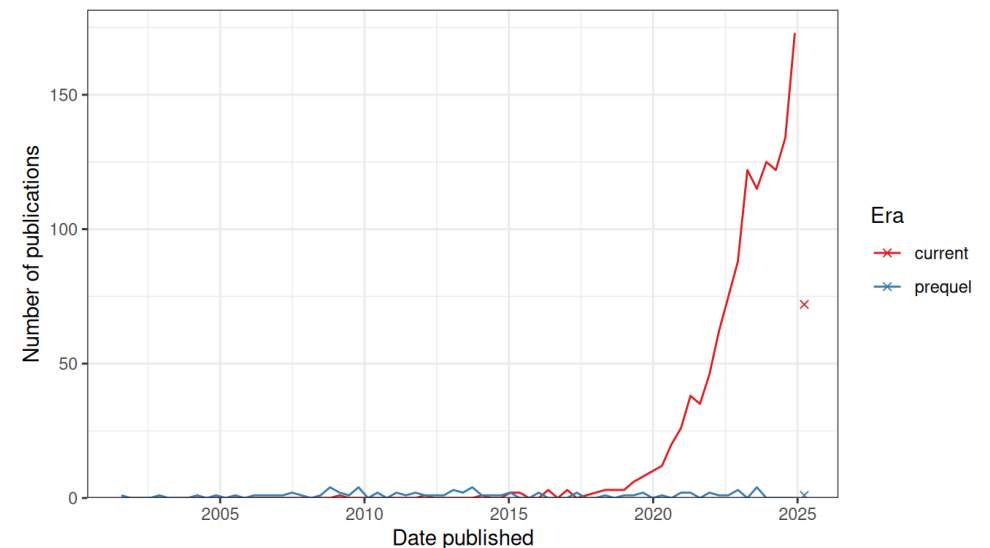
Spatial omics are widely used in cancer research, developmental biology, brain biology



Biotechnology and computational methods co-evolved in the past few years



Spatial biotechnology development and application have exponentially increased from 2020 to the present.



The corresponding analysis method has also exponentially increased from 2020 to the present.



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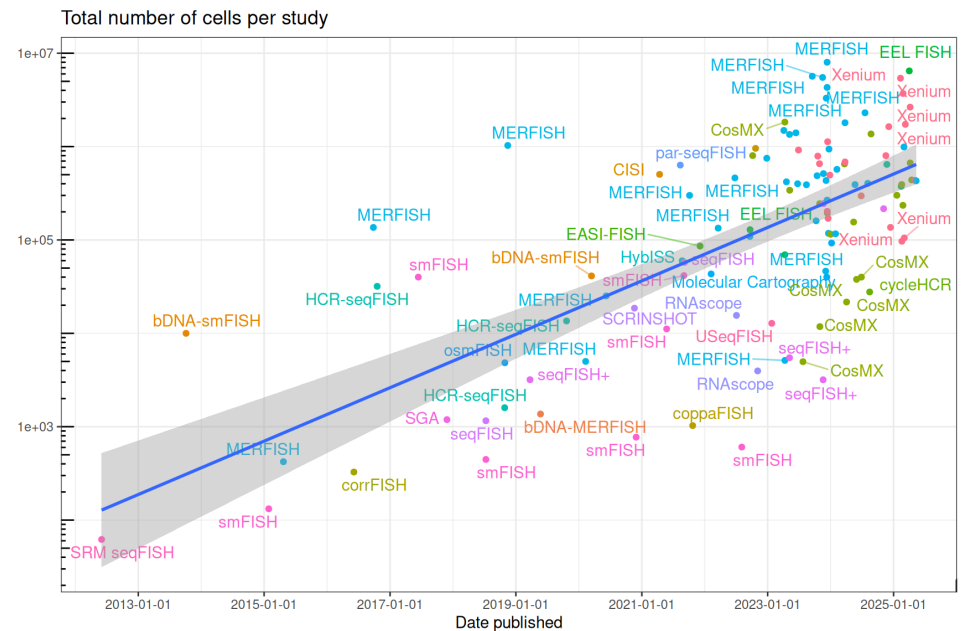
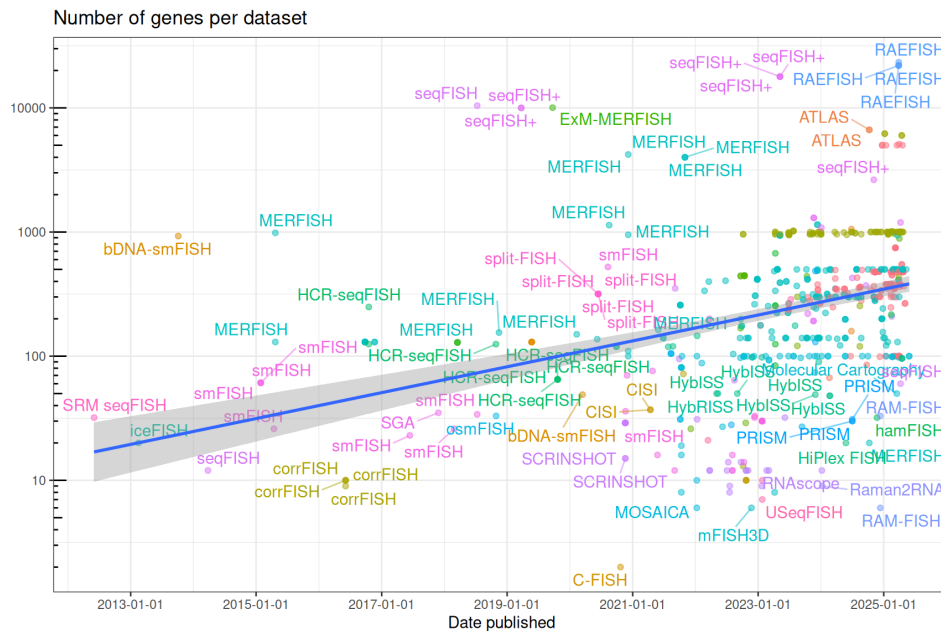
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Data and figures are from the Museum of Spatial Transcriptomics

High resolution/ high cell number/high feature number are the trend of biotechnology



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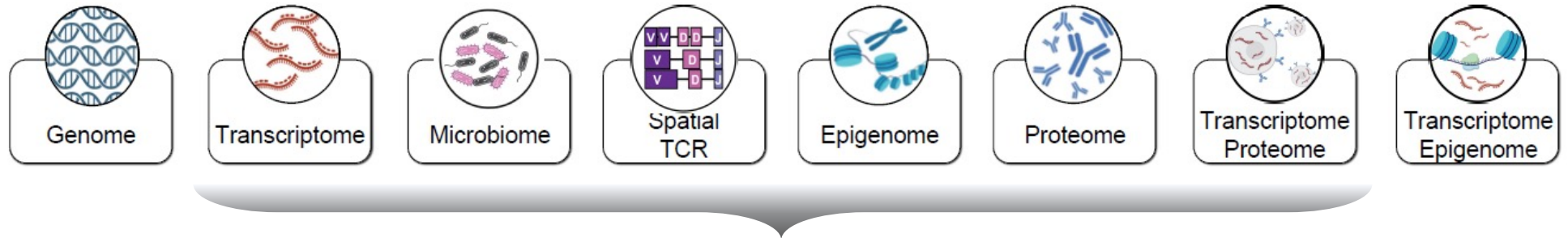
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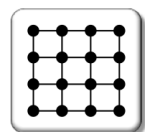
Data and figures are from the Museum
of Spatial Transcriptomics

Irregular layouts of spatial multi-omics can be naturally represented by any form of graph across multi-scale platforms



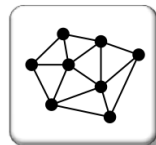
$$G = (V, E)$$

G , V , and E denote graph, node, and edge



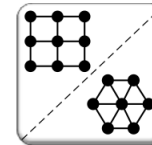
Pixel graph

Subcellular components
Node = subcellular pixel
Edge = pixel-pixel distance



Cell graph

Cellular components
Node = cell
Edge = cell-cell distance



Spot graph

Multicellular components
Node = spot
Edge = spot-spot distance



Graph signal processing enables the representation, analysis, and modification of any graph signals via graph spectral and mathematical approaches

Graph Fourier Transform is the root method for GSP

$$L = D - A$$

$$L = U\Lambda U^T$$

$$\hat{f} = U^T f$$

L : Laplacian matrix;

D : degree matrix;

A : adjacency matrix;

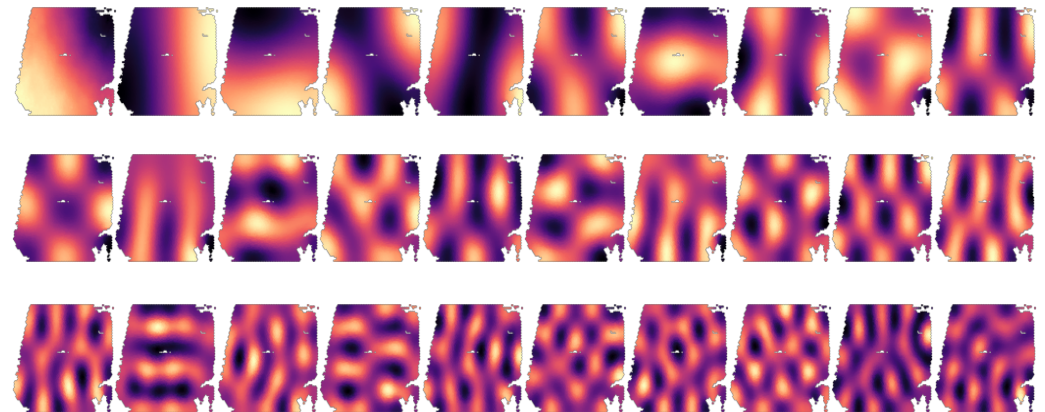
U : eigenvector (Fourier mode)

Λ : eigenvalue (Frequency)

f : graph signal

$\hat{f}$: transformed signal (Fourier coefficient)

Low frequency (small eigenvalue) of U



High frequency (large eigenvalue) of U



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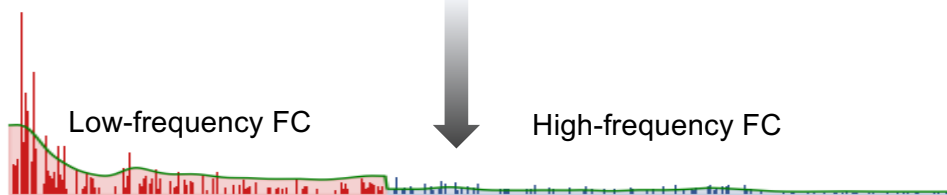
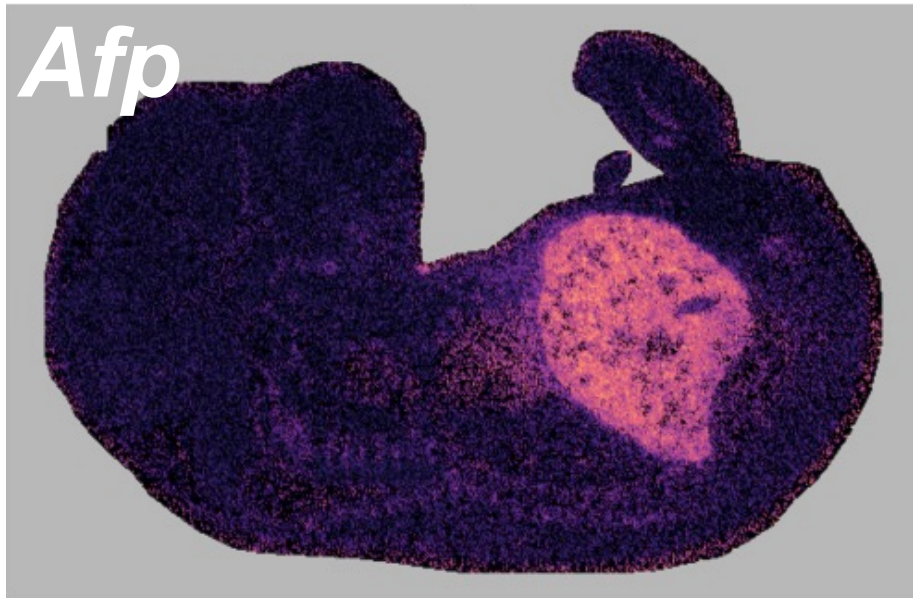
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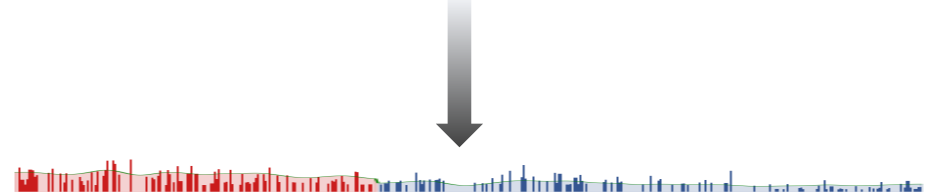
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Project signal from graph to graph spectral: an interpretable graph signal “embedding”

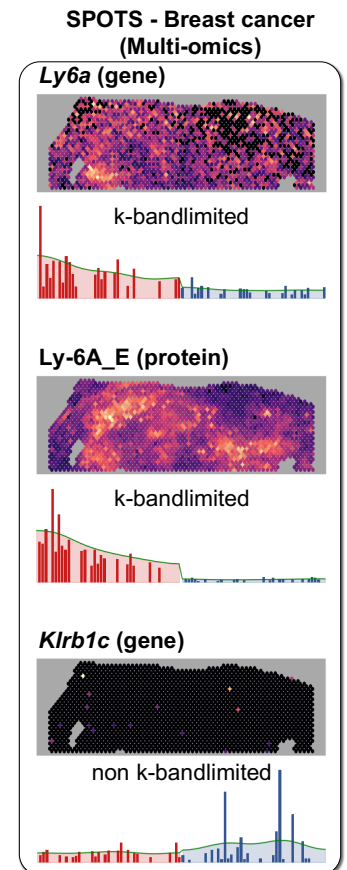
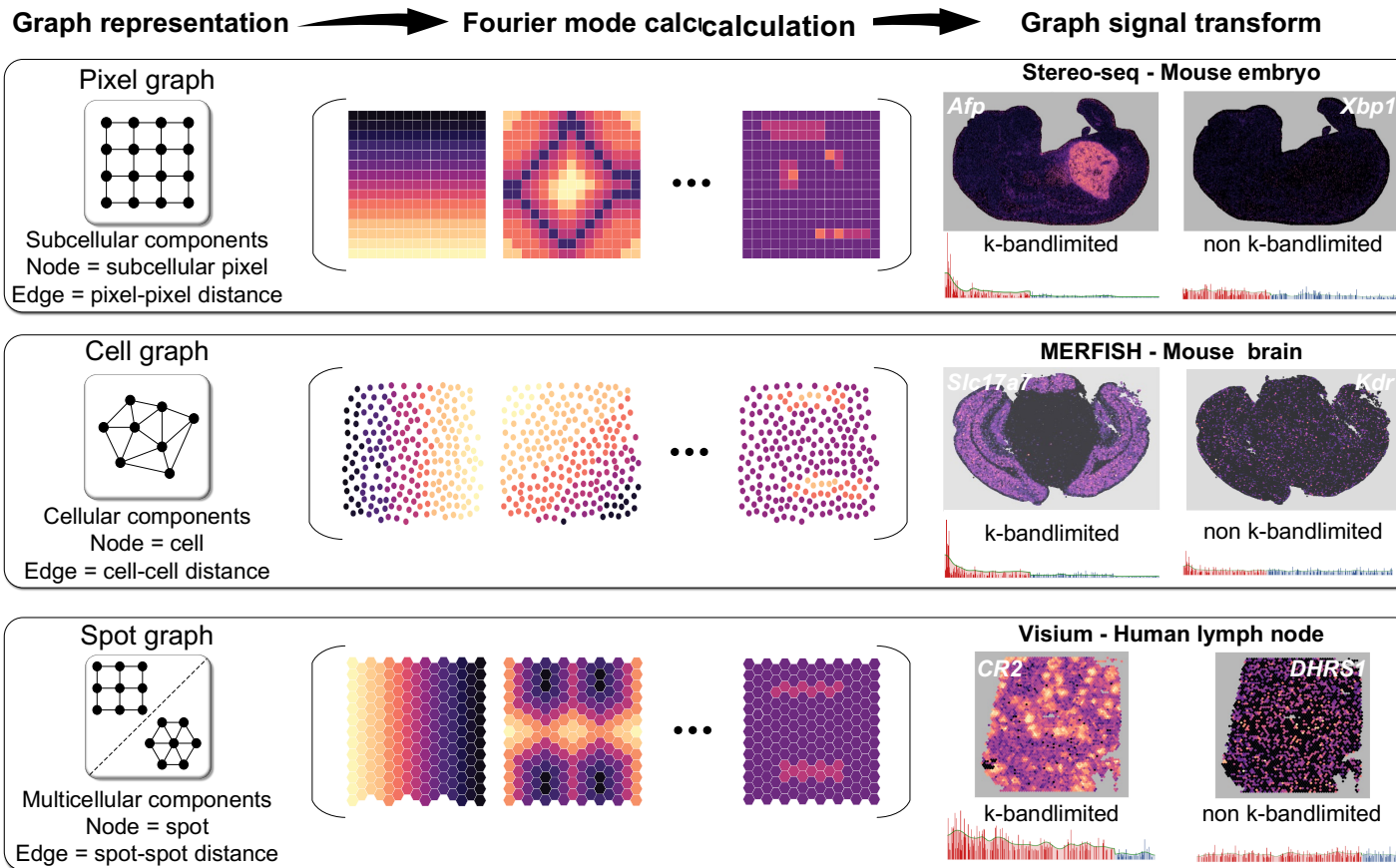
Afp: liver marker



XBP-1: regulator of endoplasmic reticulum (ER) stress

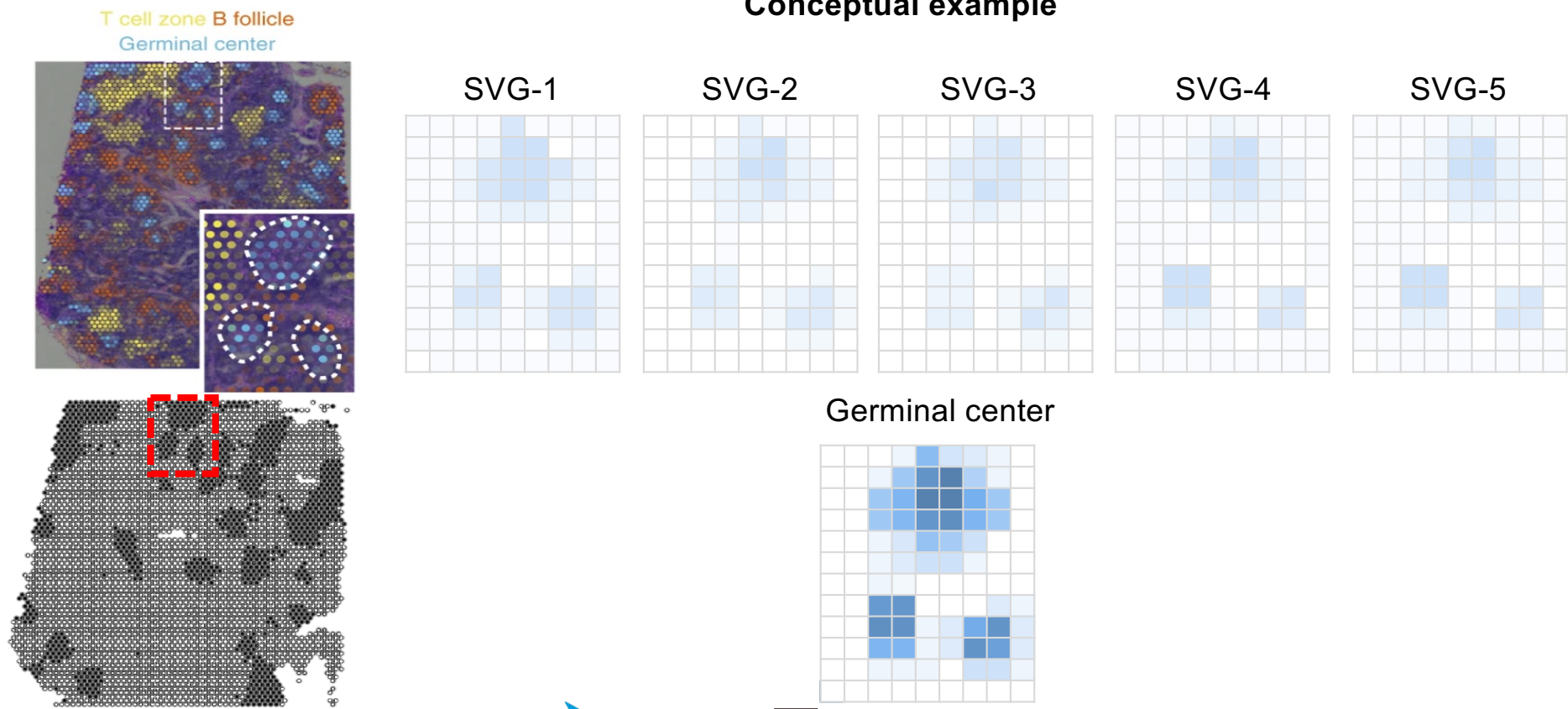


Spatial variable features can be represented and analyzed by first k-band frequencies

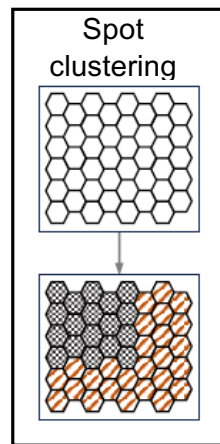


Clustering spatially variable features with similar patterns using first K-band features

Conceptual example



Add low-pass assumption to machine learning: topological regularization for multiple spatial analyses



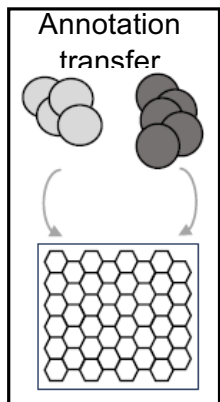
Article | [Published: 28 October 2021](#)

SpaGCN: Integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional network

[Jian Hu](#) , [Xiangjie Li](#), [Kyle Coleman](#), [Amelia Schroeder](#), [Nan Ma](#), [David J. Irwin](#), [Edward B. Lee](#), [Russell T. Shinohara](#) & [Mingyao Li](#) 

[Nature Methods](#) **18**, 1342–1351 (2021) | [Cite this article](#)

- Add topological information to features
- up to **40%** performance improvement



Article | [Open Access](#) | [Published: 16 February 2023](#)

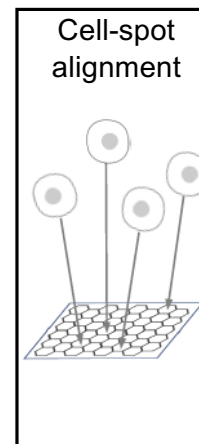
TACCO unifies annotation transfer and decomposition of cell identities for single-cell and spatial omics

[Simon Mages](#), [Noa Moriel](#), [Inbal Avraham-David](#), [Evan Murray](#), [Jan Watter](#), [Fei Chen](#), [Orit Rozenblatt-Rosen](#), [Johanna Klughammer](#) , [Aviv Regev](#)  & [Mor Nitzan](#) 

[Nature Biotechnology](#) (2023) | [Cite this article](#)

5206 Accesses | 59 Altmetric | [Metrics](#)

- Add a topological constraint to the distance matrix for Optimal transfer method
- up to **14.9%** performance improvement

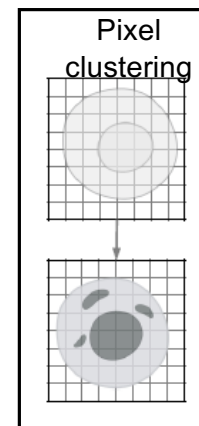


Article | [Open Access](#) | [Published: 28 October 2021](#)

Deep learning and alignment of spatially resolved single-cell transcriptomes with Tangram

[Tommaso Biancalani](#) , [Gabriele Scalia](#), [Lorenzo Buffoni](#), [Raghav Avasthi](#), [Ziqing Lu](#), [Aman Sanger](#), [Neriman Tokcan](#), [Charles R. Vanderburg](#), [Asa Segerstolpe](#), [Meng Zhang](#), [Inbal Avraham-David](#), [Sanja Vickovic](#), [Mor Nitzan](#), [Sai Ma](#), [Ayshwarya Subramanian](#), [Michal Lipinski](#), [Jason Buenrostro](#), [Nik Bear Brown](#), [Duccio Fanelli](#), [Xiaowei Zhuang](#), [Evan Z. Macosko](#) & [Aviv Regev](#) 

- Add a topological constraint to loss function
- up to **15.9%** performance improvement



Article | [Open Access](#) | [Published: 29 May 2023](#)

Learning consistent subcellular landmarks to quantify changes in multiplexed protein maps

[Hannah Spitzer](#), [Scott Berry](#), [Mark Donoghoe](#), [Lucas Pelkmans](#)  & [Fabian J. Theis](#) 

[Nature Methods](#) **20**, 1058–1069 (2023) | [Cite this article](#)

- Introduce spreading entropy to loss function
- Early convergence
- Identify **rare subcellular organelles** that are missed by the original model.



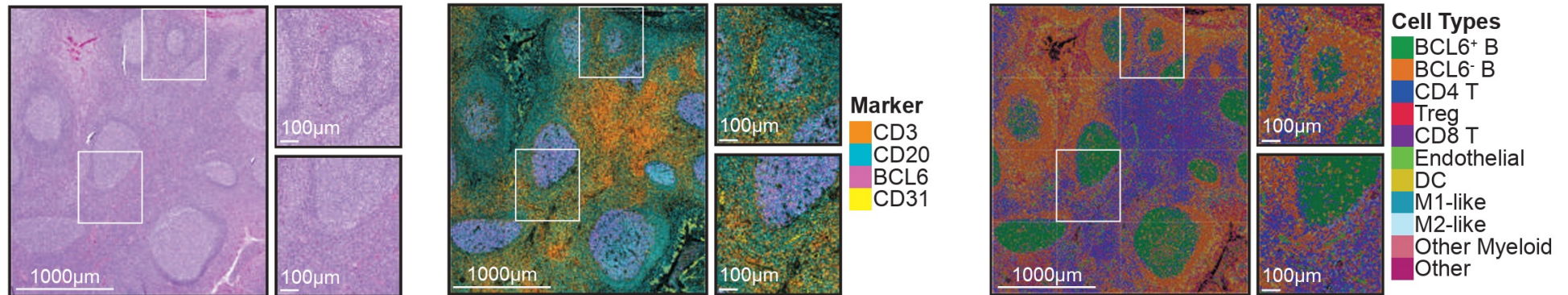
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Real data still challenges: not all features are spatially variable in complex diseases



Hematoxylin and eosin (H&E)

Multiplexed proteins

Cell phenotype determined by protein and RNA

We have:

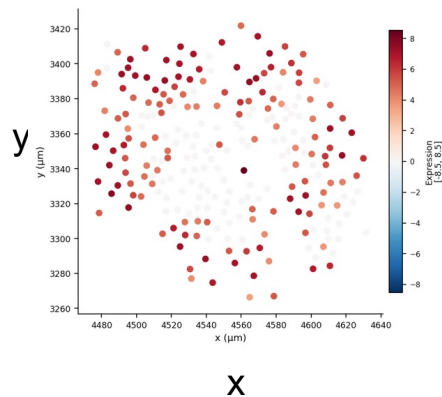
- Multi-modal analysis of a single tissue section
- Bridging histology with molecular data (protein and RNA)
- High-resolution cell atlas.
- Preservation of spatial context.

1. non-k bandlimited signal has biological functions or not has biological functions?
2. If non-k bandlimited signal has functions, how should we study non-k bandlimited signal?



Vertex / Spectrum domain views and process techniques of visiting one spatial pattern

spatial-domain view

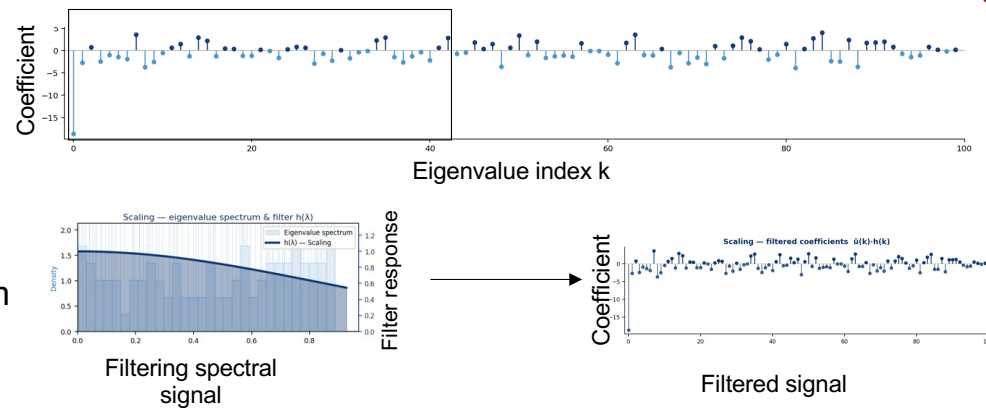


Original Spatial Gene expression

spectral-domain view

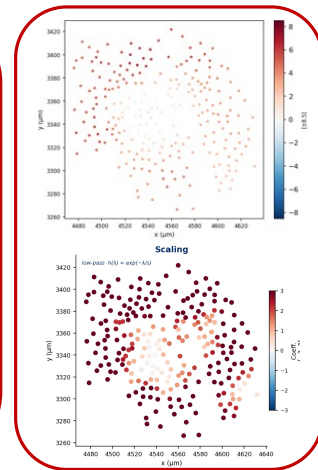
SpaGFT

Wavelet transform



Filter response: For each frequency, how much of the signal is retained.

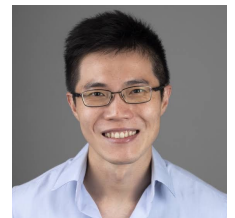
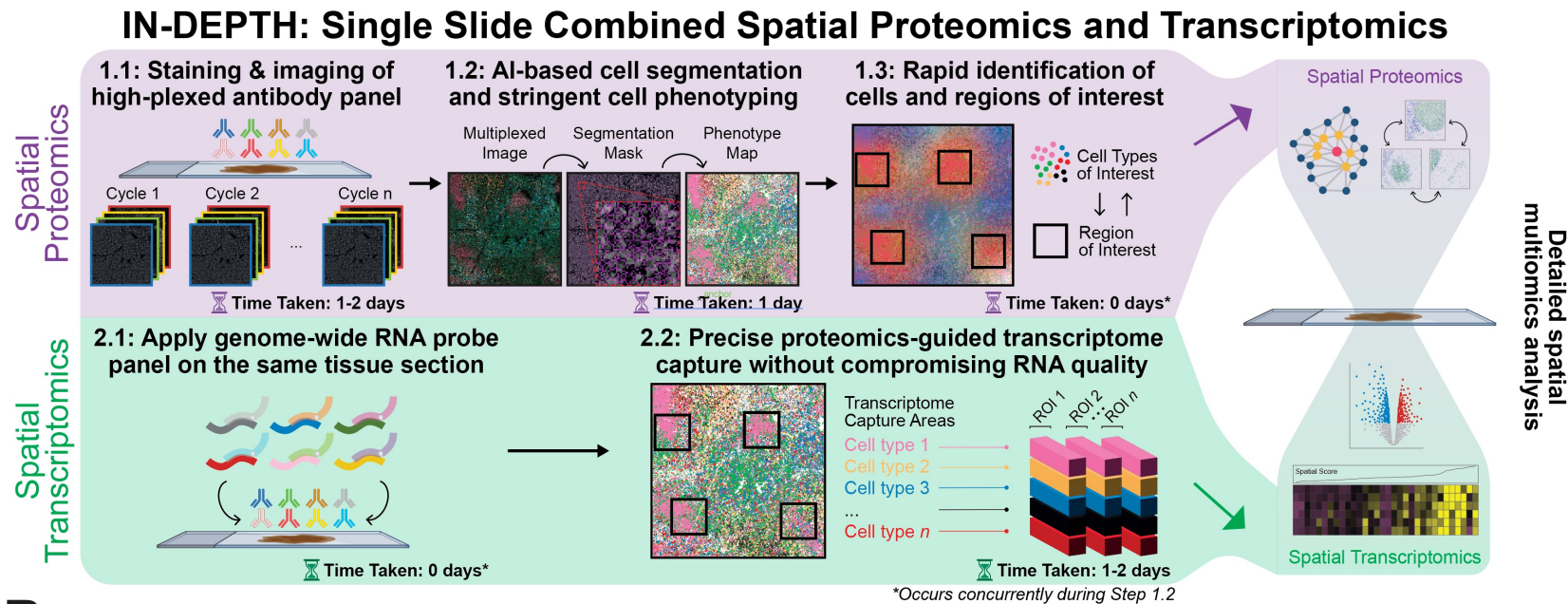
spatial-domain view



Filtered spatial gene signal



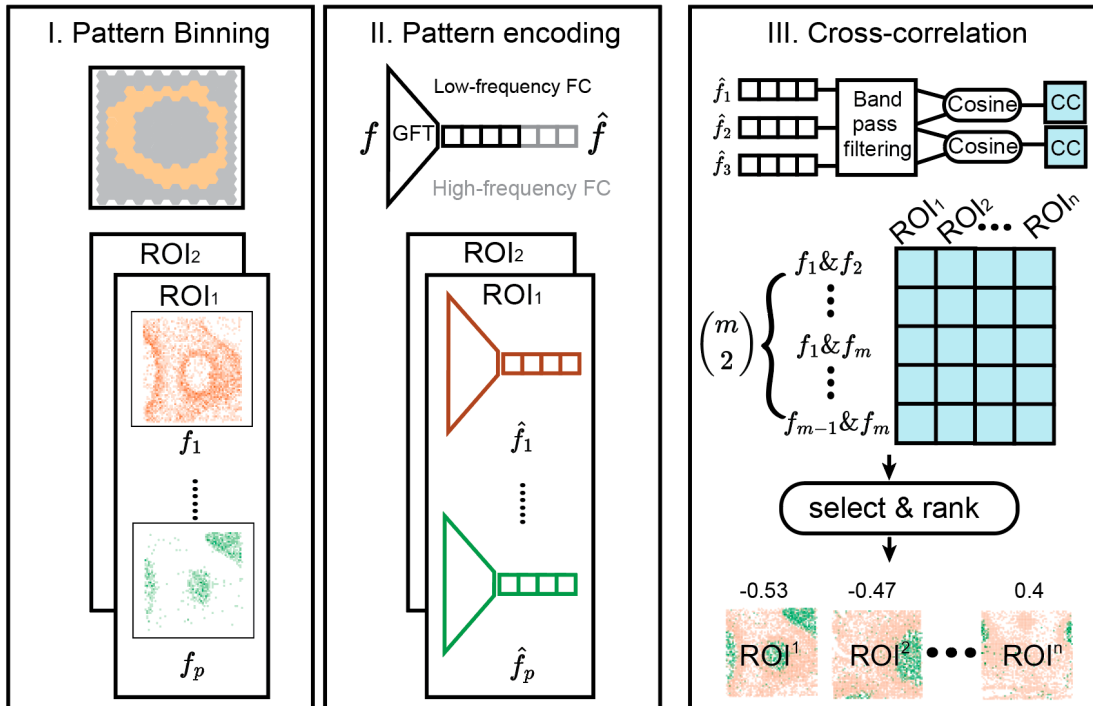
Concept adaptation to INDEPTH for spatial co-profiling of protein and RNA



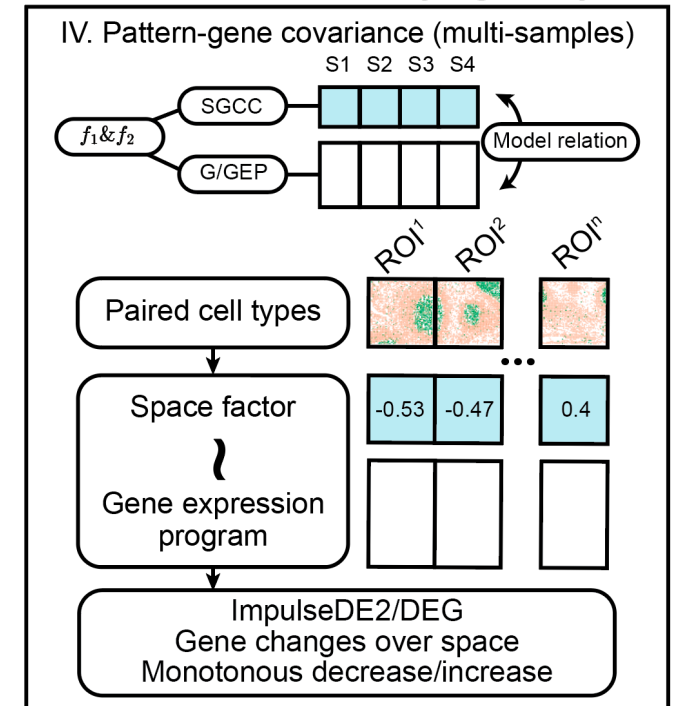
Dr. Sizun Jiang
(Beth Israel
Deaconess
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Multi-scale (i.e., band-pass) spectral graph cross-correlation defines cell pairwise relations and enables space factor-gene covariate analysis

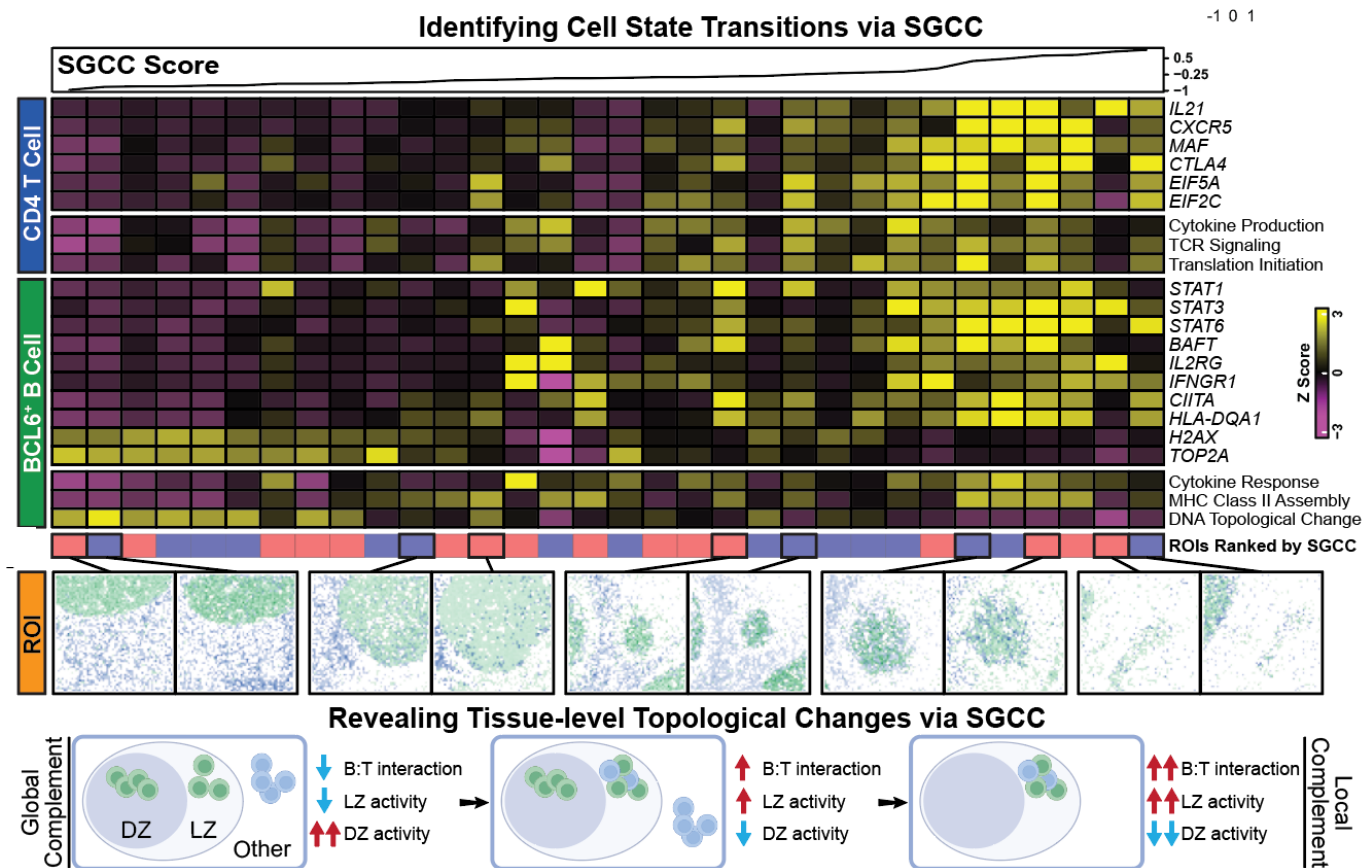
Spectral Graph Cross-Correlation (SGCC)



Space-Genes Covarying Analysis



Band-passed SGCC reveals T-B interaction across different spatial location



Acknowledgement



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Dr. Dongjun Chung



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